method	backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
single-scale:																	
FR-O [35]	R101	79.42	77.13	17.70	64.05	35.30	38.02	37.16	89.41	69.64	59.28	50.30	52.91	47.89	47.40	46.30	54.13
ICN [1]	R101-FPN	81.36	74.30	47.70	70.32	64.89	67.82	69.98	90.76	79.06	78.20	53.64	62.90	67.02	64.17	50.23	68.16
CADNet [42]	R101-FPN	87.80	82.40	49.40	**73.50**	71.10	63.50	76.60	**90.90**	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
DRN [24]	H-104	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
CenterMap [30]	R50-FPN	88.88	81.24	**53.15**	60.65	**78.62**	66.55	78.10	88.83	77.80	83.61	49.36	**66.19**	**72.10**	**72.36**	58.70	71.74
SCRDet [40]	R101-FPN	**89.98**	80.65	52.09	68.36	68.36	60.32	72.41	90.85	**87.94**	**86.86**	**65.02**	**66.68**	66.25	68.24	**65.21**	72.61
R^3Det [37]	R152-FPN	**89.49**	81.17	50.53	66.10	70.92	**78.66**	78.21	90.81	85.26	84.23	**61.81**	63.77	68.16	**69.83**	**67.17**	73.74
S^2A-Net [10]	R50-FPN	89.11	**82.84**	48.37	71.11	78.11	78.39	**87.25**	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	**74.12**
ReDet (Ours)	ReR50-ReFPN	88.79	**82.64**	**53.97**	**74.00**	**78.13**	**84.06**	**88.04**	**90.89**	**87.78**	**85.75**	61.76	60.39	**75.96**	68.07	63.59	**76.25**
multi-scale:																	
RoI Trans.* [7]	R101-FPN	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
O^2-DNet* [31]	H104	89.30	83.30	50.10	72.10	71.10	75.60	78.70	90.90	79.90	82.90	60.20	60.00	64.60	68.90	65.70	72.80
DRN* [24]	H104	89.71	82.34	47.22	64.10	76.22	74.43	85.84	90.57	86.18	84.89	57.65	61.93	69.30	69.63	58.48	73.23
Gliding Vertex* [36]	R101-FPN	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32	75.02
BBAVectors* [41]	R101	88.63	84.06	52.13	69.56	78.26	80.40	88.06	90.87	**87.23**	86.39	56.11	65.62	67.10	72.08	63.96	75.36
CenterMap* [30]	R101-FPN	**89.83**	84.41	54.60	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	**69.23**	74.13	71.56	66.06	76.03
CSL* [38]	R152-FPN	**90.25**	**85.53**	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93	76.17
SCRDet++* [39]	R152-FPN	88.68	**85.22**	54.70	73.71	71.92	**84.14**	79.39	90.82	87.04	86.02	67.90	60.86	74.52	70.76	**72.66**	76.56
S^2A-Net* [10]	R50-FPN	88.89	83.60	**57.74**	**81.95**	**79.94**	83.19	**89.11**	90.78	84.87	**87.81**	**70.30**	68.25	**78.30**	**77.01**	69.58	**79.42**
ReDet* (Ours)	ReR50-ReFPN	88.81	82.48	**60.83**	**80.82**	**78.34**	**86.06**	**88.31**	**90.87**	**88.77**	**87.03**	**68.65**	66.90	**79.26**	**79.71**	**74.67**	**80.10**

Table 6. **Comparisons with state-of-the-art methods on DOTA-v1.0 OBB Task**. H-104 means Hourglass 104. * indicates multi-scale training and testing. The results with red and blue colors indicate the best and second-best results of each column, respectively.

method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	mAP
OBB results:																	
RetinaNet-O [18]	71.43	77.64	42.12	64.65	44.53	56.79	73.31	90.84	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83	59.16
FR-O [27]	71.89	74.47	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54	62.00
Mask R-CNN [11]	76.84	73.51	49.90	57.80	51.31	71.34	79.75	90.46	74.21	66.07	46.21	70.61	63.07	64.46	57.81	9.42	62.67
HTC [2]	77.80	73.67	51.40	63.99	51.54	73.31	80.31	90.48	75.12	67.34	48.51	70.63	64.84	64.48	55.87	5.15	63.40
OWSR* [15]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	74.90
ReDet (Ours)	79.20	82.81	51.92	71.41	52.38	75.73	80.92	90.83	75.81	68.64	49.29	72.03	73.36	70.55	63.33	11.53	66.86
ReDet* (Ours)	88.51	86.45	61.23	81.20	67.60	83.65	90.00	90.86	84.30	75.33	71.49	72.06	78.32	74.73	76.10	46.98	**76.80**
HBB results:																	
RetinaNet-O [18]	71.66	77.22	48.71	65.16	49.48	69.64	79.21	90.84	77.21	61.03	47.30	68.69	67.22	74.48	46.16	5.78	62.49
FR-O [27]	71.91	71.60	50.58	61.95	51.99	71.05	80.16	90.78	77.16	67.66	47.93	69.35	69.51	74.40	60.33	5.17	63.85
HTC [2]	78.41	74.41	53.41	63.17	52.45	63.56	79.89	90.34	75.17	67.64	48.44	69.94	72.13	74.02	56.42	12.14	64.47
Mask R-CNN [11]	78.36	77.41	53.36	56.94	52.17	63.60	79.74	90.31	74.28	66.41	45.49	71.32	70.77	73.87	61.49	17.11	64.54
OWSR* [15]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	77.90
ReDet (Ours)	79.51	82.63	53.81	69.82	52.76	75.64	87.82	90.83	75.81	68.78	49.11	71.65	75.57	75.17	58.29	15.36	67.66
ReDet* (Ours)	88.68	86.57	61.93	81.20	73.71	83.59	90.06	90.86	84.30	75.56	71.55	71.86	83.93	80.38	75.62	49.55	**78.08**

Table 7. **Performance comparisons on DOTA-v1.5 test set**. Note the results of Faster R-CNN OBB (FR-O) [27], RetinaNet OBB (RetinaNet-O) [18], Mask R-CNN [11] and Hybrid Task Cascade (HTC) [2] are our re-implemented version for DOTA. OWSR [15] is a method from DOAI 2019, and we report its single model performance for fair comparisons. The HBB results of our method are converted from OBB results by calculating the axis-aligned bounding boxes. * means multi-scale training and testing.

sification accuracy due to the reduction of parameters, but it obtains higher detection mAP. We find the backbone under the cyclic group C_8 achieves better accuracy-parameter trade-off. ReResNet50+ReFPN under C_8 gains **1.83** detection mAP improvements with only **1/8** parameters (103 Mb *vs.* 12 Mb). Besides, we also extend ReResNet+ReFPN to other methods in Tab. 2. Both Faster R-CNN OBB and RetinaNet OBB with ReResNet50+ReFPN outperform its counterpart which further demonstrates the effectiveness of rotation-equivariant backbones.

Effectiveness of RiRoI Align. As shown in Tab. 3, compared with RRoI Align, RiRoI Align shows significant improvements due to its orientation alignment mechanism. While RRoI Align+MaxPool leads to a significant drop in mAP, indicating that the orientation pooling is undesirable in oriented object detection. RiRoI Align with a $l = 2$ interpolation achieves the highest **66.86** mAP and **0.87** mAP improvements than RRoI Align. Besides, we find RiRoI Align with a $l = 4$ interpolation only gains **0.33** mAP. The reason may be that too many interpolations hurt the equivariant property and inner relation between orientations.

Comparison with rotation augmentation. From another perspective, our method can be viewed as a special in-network rotation augmentation, which learns from one orientation and can be applied to multiple orientations. In contrast, rotation augmentation enhances the network by generating samples with more orientations and usually requires more time to converge. As shown in Tab. 4, although our method does not exceed the rotation augmented baseline under 1x schedule, our ReDet*, which preserves the similar amount of parameters, shows **2.59** mAP improvements with only **18%** extra training time. Moreover, the 2x baseline with rotation augmentation is **0.68** higher than our ReDet*, but it takes **twice** the training time.

Performance on other datasets. To prove the generalization of our proposed method, we also evaluate the per-

method	RC2 [19]	RRPN [22]	R^2PN [43]	RRD [16]	RoI Trans. [7]	Gliding Vertex [36]
mAP	75.7	79.08	79.6	84.3	86.2	88.2
method	R^3Det [37]	DRN [24]	CenterMap [30]	CSL [38]	S^2A-Net [10]	ReDet (Ours)
mAP	89.26	92.7*	92.8*	89.62	90.17 / 95.01*	**90.46 / 97.63***

Table 8. **Comparisons of state-of-the-art methods on HRSC2016**. * indicates that the result is evaluated under VOC2012 metrics, while other methods are all evaluated under VOC2007 metrics. We report both results for fair comparisons.

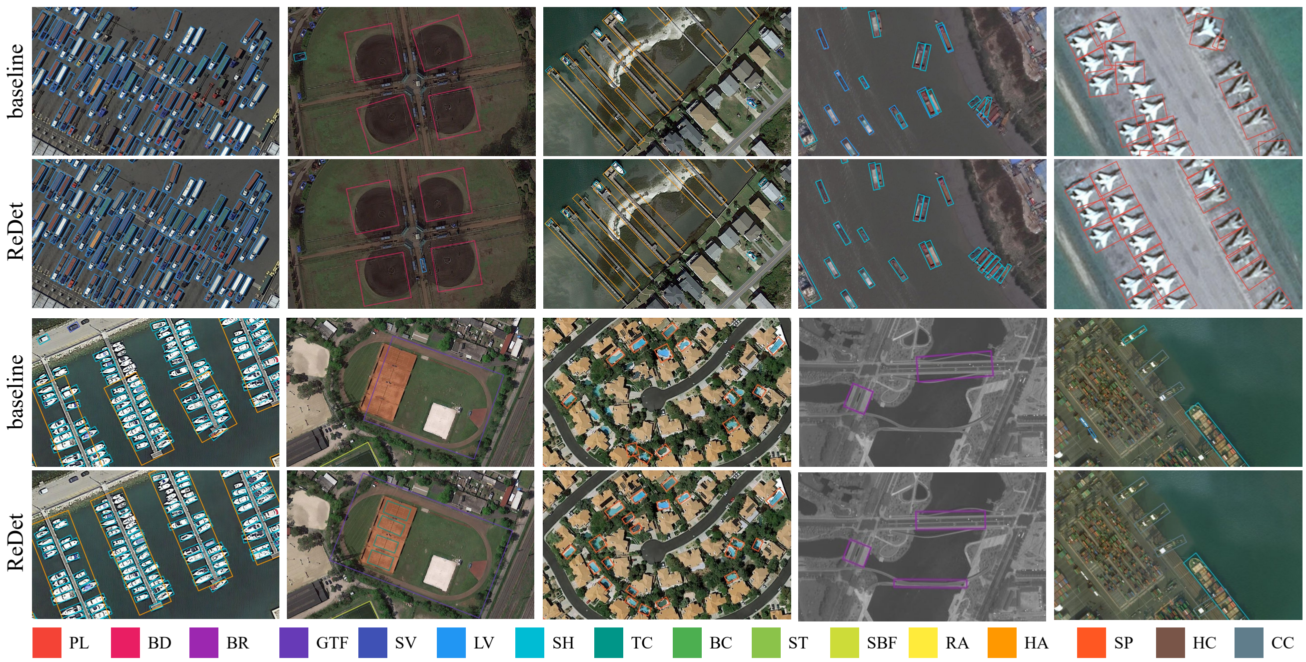

Figure 4. **Qualitative comparisons** between the proposed ReDet and the baseline method on DOTA-v1.5.

formance of ReDet on DOTA-v1.0 and HRSC2016. As is shown in Tab. 5, compared with the baseline, ReDet achieves better performance on both datasets. Moreover, ReDet has significant improvements in **AP75** and **mAP**, which demonstrates its accurate localization capabilities.

5.4. Comparisons with the State-of-the-Art

Results on DOTA-v1.0. As shown in Tab. 6, we compare our ReDet with other state-of-the-art methods on DOTA-v1.0 OBB Task. Without bells and whistles, our single-scale model achieves **76.25** mAP, outperforming all single-scale models and most multi-scale models. With limited data augmentation (*i.e.*, multi-scale data and random rotation), our method achieves state-of-the-art **80.10** mAP in the whole dataset, and obtains the best or second-best results among **12/15** categories.

Results on DOTA-v1.5. Compared with DOTA-v1.0, DOTA-v1.5 contains many extremely small instances, which increases the difficulty of object detection. We report both OBB and HBB results on DOTA-v1.5 test set in Tab. 7. With single-scale data, our method achieves **66.86** OBB mAP and **67.66** HBB mAP, outperforming RetinaNet OBB, Faster R-CNN OBB, Mask R-CNN [11] and HTC [2] by a large margin. Especially for the categories with small instances (*e.g.*, HA, SP, CC) and large scale variations (*e.g.*, PL, BD), our method performs better. Besides, as shown in Fig. 2, our ReDet achieves better parameter *vs.* accuracy trade-off, which further demonstrates its efficiency. Compared to previous best results by OWSR [15], our multi-scale model achieves state-of-the-art performance, about **76.80** OBB mAP and **78.08** HBB mAP. Qualitative comparisons between our ReDet and the baseline method are visualized in Fig. 4.

Results on HRSC2016. The HRSC2016 contains a lot of thin and long ship instances with arbitrary orientation. We compare our ReDet with other state-of-the-art methods in Tab. 8. Our method achieves the state-of-the-art performance, *i.e.*, with mAP of **90.46** and **97.63** under the VOC2007 and VOC2012 metrics, respectively.

6. Conclusions

This paper presents a Rotation-equivariant Detector for aerial object detection, which consists of two parts: the rotation-equivariant backbone and the RiRoI Align. The former produces rotation-equivariant features, while the latter extracts rotation-invariant features from rotation-equivariant features. Extensive experiments on DOTA and HRSC2016 demonstrate the effectiveness of our method.

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